DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2025-26
PRE-BOARDS EXAMINATION

CLASS: X | **FULL MARKS: 100**
SUBJECT: COMPUTER APPLICATIONS [SET A] | **TIME: 2 HOURS**

- ***Answers to this Paper must be written on the paper provided separately.***
- ***You will not be allowed to write during the first 15 minutes.***
- ***This time is to be spent in reading the question paper.***
- ***The time given at the head of this Paper is the time allowed for writing the answers.***
- ***This paper consists of six printed pages. This Paper is divided into two Sections.***
- ***Attempt all questions from Section A and any four questions from Section B.***
- ***The intended marks for questions or parts of questions are given in brackets [].***

SECTION A(40 Marks)
(Attempt all questions from this Section)

Question 1 **[20x1=20]**

Choose the correct answers to the questions from the given options

(i)

Identify the feature of java that resembles job of the delivery depicted by the above picture:

(a) Call by value (b) Method overloading (c) Call by reference (d) Class and object

(ii) **Andrea wants to store 5 values in an array. Help her to choose the correct option to accomplish the same.**

i.for (i=0;i<5;i++)
ii.int i;
iii.arr[i]=sc.nextInt( );
iv.int arr[ ]=new int[5];

(a) i,iv,ii,iii (b) ii,iv,i,iii (c) iii,ii,i,iv (d) ii,iv,iii,i

(iii) **Choose the correct output:**
System.out.println(Math.sqrt(Math.pow(Math.max(3,1,2),2)));

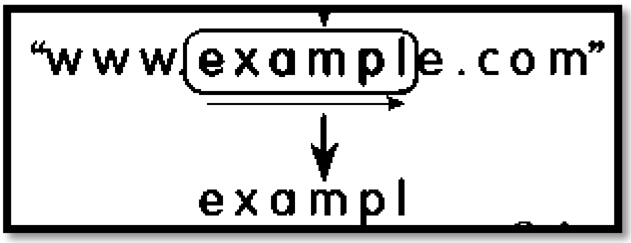
(a) 9.0 (b) 3.0 (c) Error (d) 3

(iv) **A double primitive type consumes _______ of memory.**

(a) 2bytes (b)4bytes (c) 16bytes (d) 8bytes

(v) **Choose the correct output:**
System.out.println(String.valueOf(9875).substring(2,2));
(a) Error **(b) Null** **(c) 8** **(d) -1**

(vi) **Identify the statement that justifies the given image**



(a) "www.example.com".substring(4,11)
(b) "www.example.com".substring(4,"www.example.com".length()-2)
(c) "www.example.com".substring(4,9)
(d) "www.example.com".substring(4,"www.example.com".length()-5)

(vii) **Choose the correct option :**
System.out.println(String.valueOf("123")+Integer.parseInt("456"));
(a) 579 **(b)123456** **(c)Error** **(d) 456123**

(viii) **Choose the correct output if n=3:**
```
int i = 1, s = 0;
for (; i <= n; ) {
 s += i;
 i++;          }
 System.out.println(s);
```
(a) 6 **(b) Error** **(c)15**
(d) value of s will be printed 3times

(ix) **Choose the correct output:**
String s1 = "Hello";
String s2 = "HELLO";
System.out.println(s1.equalsIgnoreCase(s2));
(a) true **(b) false** **(c)Hello** **(d)Error**

(x) **Assertion(A): A method declared as static can be called using the class name without creating an object.**
Reason(R): Static methods belong to the class rather than an instance of the class.
(a) Both Assertion (A) and Reason (R) are true and (R) is a correct explanation of (A)
(b) Both Assertion (A) and Reason (R) are true and (R) is not a correct explanation of (A)
(c) Assertion (A) is true and Reason (R) is false
(d) Assertion (A) is false and Reason(R) is true

(xi) **Which of these symbol is a punctuator (separator) in Java?**
(a) { } **(b) +** **(c) &&** **(d) =**

(xii) **What is the output of the following code?**

```
double d = 9.78;
int i = (int)d;
System.out.println(i);
```

(a) 10 **(b) 9.78** **(c) 9** **(d) Error**

(xiii) **Assertion(A): The size of an array can be changed after it is created.**

Reason(R): Java allows dynamic memory allocation for arrays.

(a) Both Assertion (A) and Reason (R) are true and (R) is a correct explanation of (A)

(b) Both Assertion (A) and Reason (R) are true and (R) is not a correct explanation of (A)

(c) Assertion (A) is true and Reason (R) is false

(d) Assertion (A) is false and Reason(R) is true

(xiv)

22	33	44	55
12	34	56	78
10	5	20	11

A Double Dimensional Array named as Mat of size 3x4 is given. Choose the correct option based on the given question:

What will be the value of val, if val= Mat[1][2]+Mat[0+1][1+2]-Mat[4-2][4-4]

(a) Error **(b) 124** **(c) 241** **(d) Exception**

(xv) **The given structure is an example of ____________________**

```
class Name
{ }
```

(a) Encapsulation **(b) Package** **(c) Inheritance** **(d) Polymorphism**

(xvi) **A method that modifies the state of the program or produces side effects outside its own scope.**

(a) Pure Method **(b) Impure Method** **(c) Method Overloading**

(d) Method Prototype

(xvii) **Which kind of type casting is also called narrowing conversion?**

(a) int → double **(b) double → int** **(c) byte → short** **(d) float → double**

(*xviii*) **Identify the correct output:**

```
class qwerty {
static int a=10;
int b=5;
static void main()   {
qwerty q1=new qwerty();
qwerty q2=new qwerty();
int q3 = q1.a+q2.b;
System.out.println(q3);  }}
```

(a) 5 **(b) 10** **(c) 15** **(d) Error**

(xix) **Which of the following is a valid numeric literal in Java?**

(a) 17 (b) AB9 (c) 7.0.1 (d) 7,001

(xx) **Which of the following is a blueprint for creating objects in Java?**

(a) Object (b) Class (c) Method (d) Constructor

Question 2

i. **Write the Java expression for :** [2]

$$\frac{a^x + b^y}{\sqrt[3]{x} + \sqrt[3]{y}}$$

ii. **Evaluate the expression: s+=s-- + ++t/t + t--; where t=10,s=15;** [2]

iii. **How many times will the following loop execute? What value will be returned?** [2]

```
int meth( ) {
int x=2; int y=50;
do{
     ++x;
      y-=x++;
     }while(x<=10);
return y;}
```

iv. **Consider the array:** [2]

int a[]={12,35,40,42,56,59,70};

(a) In the above given array, using binary search, how many iterations are required to check for the existence of the value 56?

(b) If the array is arranged in descending order, how many iterations are required to check for the existence of 56 using linear search?

v. **Identify the errors and rewrite the correct code:** [4]

```
class Sample
      { public static void main( )
      { int n,p; float k,r; n=-25; p=-12;
        if(n= 25)
      { k=Math.sqrt(p);
        System.out.println("The value of"+p+"=" +k); }
        else
      { r=Math.cbrt(n);
        System.out.println("The value of"+n+"="+r); } } }
```

vi. **Predict the output of the following program segment:** [2]

```
class PQR
     { static int x=7;   int y=2;
       public static void main( )
    {  PQR obj1=new PQR( );
       PQR obj2=new PQR( );
       obj1.x=10;
       obj1.y=20;
```

```
        System.out.println(obj2.x);
        System.out.println(obj2.y); }    }
```

vii. **Predict the output of the following program segment:** **[2]**

```
int x[ ][ ]={{4,3,7},{8,9,10},{11,12,13}};
int p=x[1][1]% x[0][0]+ ++x[2][0]* x[1][2]/x[2][1]++;
int q=x.length;
System.out.println("p="+p);
System.out.println("q="+q);
```

viii. **With the help of an example show how constructors are different from method.** **[2]**

ix. **Give differences between length and length() with the help of an example.** **[2]**

SECTION B (60 Marks)

(Answer any four questions from this Section)

The answers in this section should consist of the programs in either BlueJ environment or any program environment with java as the base. Each program should be written using variable description/mnemonic codes so that the logic of the program is clearly depicted. Flowcharts and algorithms are not required.

Question 3 [15]

Define a class named Plot with the following description:

Class Name : Plot

Member variables :

double a : to store the area of the plot

double price : to store the total cost of the plot

double regis : to calculate the registration amount

Member functions

Plot( ) : Default constructor to initialize data members

void input( ) : to accept the area plot

void pay () :

area	price	Registration on the total cost
<=100 sq meter	5 lakhs	2%
Next 500 sq meter	3 lakhs/ sq meter	5%
Next 500 sq meter	2 lakhs /sq meter	7%
Above 1100 sq meter	1 lakh/ sq meter	8%

void show( ) : To display the area, price and registration amount of the plot after discount in the given format:

Area Total Cost Registration Amount

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Define a class with the above mentioned specifications, create the main method, create an object and invoke the member methods.

Question 4 [15]

Write a program to accept a range m and n, where m<n. Display all the Smith numbers from the given range.
A Smith number is a composite number where the sum of its digits equals the sum of the digits of its prime factors (excluding 1).
Example: 22 is a Smith Number as the sum of the digits are: 2+2=4. The prime factors of 22 are: 2 and 11 , Sum = 2+(1+1) = 4

Question 5 [15]

Design a class to overload a function fun( ) as follows:
void fun(String a[ ]): to display those words from the array that endswith 'P'.

void fun( ): dipslay the following pattern.
B L U E J
B L U E
B L U
B L
B

void fun(int n): to generate tribonacci series of n terms.
The tribonacci series is a generalization of the Fibonacci sequence where each term is the sum of the three preceding terms.
a(n) = a(n-1) + a(n-2) + a(n-3) with a(0) = a(1) = 0, a(2) = 1.
First few numbers in the Tribonacci Sequence are 0, 0, 1, 1, 2, 4, 7, 13, 24, 44, 81, 149, 274, 504, ...

[Do not write the main( ) method]

Question 6 [15]

Write a program to accept a sentence and check if it is a pangram or not.
If it is a pangram then display total number of consonants and vowels from the sentences otherwise display "Not a pangram".
[A pangram is a sentence or phrase that contains every letter of a given alphabet at least once. The most famous example in English is "The quick brown fox jumps over the lazy dog," which uses all 26 letters.]

Question 7 [15]

Write a program to accept size 'n' of a Single Dimensional Array and create three different arrays to store Name, contact number and City the liv of 'n' customers. Use selection sort to sort the arrays w.r.t the name and display the sorted arrays as per the given format:
Example:
n=4
Before Sorting
Array1[ ]={"Jack ", "Arita", " Sonam", " Ranjit"}
Array2[ ]={9089786756, 9988776655, 9897969594, 8679452345}
Array3[ ]={"Delhi ", " Jaipur", "Chennai", "Kolkata" }

After Sorting
Array1[ ]={ "Arita", "Jack ", " Ranjit", " Sonam" }
Array2[ ]={ 9988776655, 9089786756, 8679452345, 9897969594}

Array3[ ]={ " Jaipur" ,"Delhi ", "Kolkata" , "Chennai"}

Question 8 [15]

Write a program to accept the size (m x m), where m is the number of rows and columns and numbers from the users in a Double Dimensional Array and perform the following:

i. **Display the original array**
ii. **Display the Boundary elements.**
iii. **Display the sum of Non-diagonal elements.**